DIGITAL EXISTENCE AS ONTOLOGICAL INTERVENTION: BACKSTAGE NETWORKS IN POSTWAR TAIWANESE POPULAR MUSIC

Ming Cheng[1] **Yung-Chuan Chang**[1] **Chien-Chang Yang**[2] **Li Su**[1]
[1] Institute of Information Science, Academia Sinica, Taipei, Taiwan
[2] Graduate Institute of Musicology, National Taiwan University, Taipei, Taiwan
hugowski@iis.sinica.edu.tw, yungchuan@iis.sinica.edu.tw,
chienchang@ntu.edu.tw, lisu@iis.sinica.edu.tw

ABSTRACT

The historical narrative of popular music privileges front-stage stars, and music information retrieval (MIR) has largely reproduced this bias. Backstage laborers—arrangers, session musicians, engineers—remain absent from the field's datasets and research questions. Responding to Born's (2020) call for a critical MIR, we argue that encoding historically marginalized music workers into computational databases constitutes an *ontological intervention* that grants them *digital existence* in an era where absence from metadata means erasure from history. We introduce the Private Archive of Scanned Music Artifacts (PASMA), a dataset reconstructed by an LLM pipeline from 1,122 Taiwanese vinyl liner notes spanning 1957–1991, and combine Social Network Analysis with MERT embeddings on vocal-separated stems. The collaboration network transformed sharply around 1978, when multitrack recording shifted it from a fragmented, high-modularity ecology to a centralized giant component bridged by elite intermediaries. MERT embeddings indicate that the arrangement traditions defined by these network communities occupy distinct regions in acoustic space. These findings demonstrate that MIR's computational tools, redirected toward the backstage, can decode historical power structures and restore the visibility of the laborers who built the soundscapes of postwar Taiwan.

1. INTRODUCTION

The historical narrative of popular music has long been organized around front-stage figures—stars, singers, and songwriters. This emphasis is not merely a cultural habit; it has profoundly shaped the computational study of music. In music information retrieval (MIR), audio features and metadata are predominantly mobilized for consumer-facing applications: recommendation systems match listeners to products via the names of credited performers [1], while hit prediction models treat chart success as a property of songs or artists [2]. Yet recorded music is not the product of individual genius. It is a collective technical and artistic practice [3], assembled by extensive networks of backstage laborers—arrangers, session musicians, recording engineers, and producers—working within the material constraints of specific studios and technologies [4, 5]. These workers remain largely invisible in MIR research, not because they are unimportant, but because the data infrastructures and research questions of the field have never been designed to see them.

Technically, metadata on backstage contributors has never been systematically digitized: liner-note credits—where such information was recorded—remain locked in physical artifacts, and their digitization is among the most labor-intensive tasks in music data curation. Conceptually, however, the problem runs deeper. We have tended to treat recording technology as a neutral tool and backstage workers as mere executors—interchangeable hands operating equipment in service of a composer's or performer's vision. This technological determinist view naturally obscures the role of backstage workers as agents who negotiate and reshape technology through practice [5, 6]. As Born argues in her call for a critical and diversified MIR, the field's dominant knowledge representations systematically privilege certain musical ontologies over others, reproducing the very inequalities that structure the industries themselves [7]. This structural condition has also been documented empirically: an informetric study of ISMIR proceedings from 2000–2015 found that only 14.1% of papers were led by female researchers, with little improvement over time [8].

Born's critique raises a question that extends beyond the politics of representation within MIR. In her analysis, she warns that the reduction of music to computational knowledge representations—classes, attributes, relations—risks conflating epistemology with ontology, flattening the embodied and social existence of music into mere data [7]. We take this warning seriously, but argue for a reversal. For historically marginalized subjects such as the backstage session musicians in our study, the situation is not one of *reduction* but of *absence*: they have never been encoded in any database at all. Absence from metadata is therefore tantamount to erasure from the historical record. For these workers, being inscribed into the very "epistemological"



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infrastructures that Born critiques is precisely what grants them *digital existence*—a concrete presence in the computational record from which they have been excluded. Our study therefore appropriates MIR's knowledge representation tools not as instruments of reduction, but as a means of *ontological intervention*: restoring the historical existence of hidden laborers by making their collaborative ties visible and analyzable for the first time.

We ground this argument in Taiwan's popular music industry between 1957 and 1991. This period spans the island's most consequential socio-technical transformation in recording: from an industry shaped by colonial Taiwanese song and Shanghai Mandarin pop to the industrial adoption and standardization of multitrack studio workflows in the late 1970s [9]. Viewed through the lens of the Social Construction of Technology (SCOT) [6], this transformation fundamentally reorganized who could participate in production, how labor was divided, and where aesthetic authority resided. To investigate this transformation, we introduce the Private Archive of Scanned Music Artifacts (PASMA) dataset, reconstructed from the liner notes of 1,122 vinyl records through a multi-stage pipeline assisted by Large Language Models (LLMs) for OCR correction, semantic parsing, cross-checking release dates, and role classification and correction. The extracted credits are modeled as a time-varying collaboration network and analyzed through Social Network Analysis (SNA); vocal-removed instrumental stems are processed by MERT-330M [10], an acoustic music understanding model pretrained with large-scale self-supervised learning, to extract embeddings of backstage acoustic signatures, separated from front-stage vocal performance. Our results reveal a sharp structural transformation around 1978: the network shifted from a fragmented, high-modularity ecology of independent communities to a single giant component centered on recording studios and bridged by arrangers and session musicians. MERT embeddings indicate that the arrangement traditions defined by these communities occupy distinct regions in acoustic space, suggesting that the reorganization of social ties is entangled with measurable changes in musical sound. Together, these findings demonstrate that MIR's computational tools, redirected toward historically invisible subjects, can function as instruments of ontological intervention—granting backstage laborers a digital existence that conventional narratives have denied them.

2. BACKGROUND AND THEORETICAL FRAMEWORK

2.1 Born's Critique and Agonistic Interdisciplinarity

Born's intervention goes beyond a call for topical diversity in MIR. She identifies a structural condition: the field's institutional composition, overwhelmingly drawn from computer science, has naturalized research agendas shaped by the commercial digital music economy, while the datasets and benchmarks that anchor MIR research encode particular assumptions about what music is, who makes it, and which aspects of music-making merit representation [7]. Her proposed remedy is *agonistic interdisciplinarity*—a mode of engagement in which computational and humanistic disciplines interact as equals, each capable of transforming the other's methods and assumptions. Our study attempts to instantiate this model. The research question originates in music history and labor studies; the computational tools—SNA and audio embeddings—reveal patterns inaccessible through archival reading alone; and the historical framing prevents those patterns from collapsing into decontextualized pattern recognition.

2.2 SCOT, Cultural Intermediaries, and Studio Power

Our technical analysis is grounded in the Social Construction of Technology (SCOT) framework [6], and particularly in Horning's demonstration that developments in audio technology were frequently driven by feedback from studio users, establishing a co-construction of technology and studio practice [5]. We adopt this anti-determinist stance to interpret the industrial adoption and standardization of multitrack studio workflows in late-70s Taiwan: not as a neutral upgrade, but as an artifact whose meaning was actively shaped by the backstage workers who adopted it.

Prior to the late 1970s, mono or limited-track synchronous methods demanded real-time coordination among musicians sharing the same physical space. Multitrack recording dissolved these constraints through overdubbing, granting arrangers and engineers unprecedented control over the final sound. To describe the sociological significance of this shift, we draw on Bourdieu's concept of *cultural intermediaries* [11]: recording professionals who do not merely execute technical tasks, but function as gatekeepers negotiating the aesthetic rules of production. As multitrack technology enabled capital-intensive, studio-centered workflows, a small group of intermediaries consolidated control over the means of production.

2.3 Network Approaches in MIR and Musicology

Network science in MIR has largely served commercial or classificatory ends: tripartite graphs for hit prediction [2], community detection on audio-feature similarity [12], and centrality-based analysis of sampling influence among front-stage artists [13]. A separate line of work, exemplified by the Polifonia project, addresses musical heritage through ontology networks and knowledge graphs, explicitly recognizing that existing models are insufficient for representing music history and cultural context [14]. Closer to actual social relations, McAndrew and Everett applied SNA to elite British classical composers [15], while qualitative studies have long recognized the creative agency of uncredited session musicians [4]. Our study bridges these traditions: we appropriate network tools not to cluster audio or trace front-stage lineage, but to reconstruct concrete historical networks of the backstage laborers that qualitative musicology has described, and to link their social ties to acoustic outcomes at scale.

3. DATASET AND METHODOLOGY

3.1 Data Reconstruction: The PASMA Dataset

The foundation of this study is the PASMA dataset, reconstructed from the liner notes of 1,122 Taiwanese vinyl records issued between 1957 and 1991. Liner notes from this period credited backstage workers, including arrangers, session musicians, recording engineers, and producers, with varying but often remarkable detail. They therefore constitute an archival source for reconstructing the social organization of recording work, but one that has never been systematically digitized.

In this study, a "role" refers to the credited function that a person or institution performed in relation to a specific album, such as arrangement, recording, performance, production, design, or release. Figure 1 shows our LLM-assisted pipeline, which comprises five stages. First, OCR is performed on scanned liner-note images whose heterogeneous typography combines multiple scripts and orientations; because traditional OCR engines perform poorly on such layouts, raw output is post-corrected by an LLM that resolves character-recognition errors using surrounding linguistic context. Second, the corrected text—unstructured free-form prose rather than tabular metadata—is parsed by an LLM to extract triples of (person or institution, role, album). Third, release years are cross-checked across multiple sources to resolve inconsistencies. Fourth, entity disambiguation reconciles variant spellings, pen names, and inconsistent transliterations across multiple scripts—the most error-prone step in the pipeline. Finally, heterogeneous role labels are classified into a controlled vocabulary of 15 person-role and 5 institution-type categories. The resulting dataset contains 3,688 unique persons and institutions, each linked to albums with classified roles and date metadata. We plan to release a research version with personal and institutional names replaced by stable identifiers.

Validation followed an iterative, expert-in-the-loop procedure rather than a single hold-out evaluation: because the historical credits PASMA represents have no external ground truth against which automated accuracy can be computed, we instead validated the pipeline through repeated cycles of network visualization and correction, with anomalies detected by domain expertise traced back to their source records and resolved. This procedure leverages extensive professional experience in Taiwan's recording industry held by one of the authors. The trade-off is uneven coverage—high-degree and bridging nodes were densely scrutinized while peripheral nodes received less attention—an asymmetry we accept because errors at high-degree nodes would most distort the structural findings.

3.2 Social Network Analysis

The extracted credit data is modeled as a time-varying bipartite graph in which persons and institutions are linked to the albums they are credited on [16]. For each five-year sliding window (one-year step) across 1957–1991, the bipartite graph is projected into a unipartite collaboration network: two persons share an edge if they are credited on the same album within the window, and the edge is weighted by the number of shared albums. We then compute four diagnostic metrics on each windowed network: the ratio of the largest connected component (global integration), the average clustering coefficient [17] (local triadic closure), Louvain modularity [18] (meso-level partition strength), and Burt's structural hole constraint [19] (bridging position at the node level). The first three characterize how the industry as a whole consolidated over time; the last identifies the specific individuals and institutions—the cultural intermediaries—whose structural positions carry direct implications for the acoustic character of the music produced within their networks.

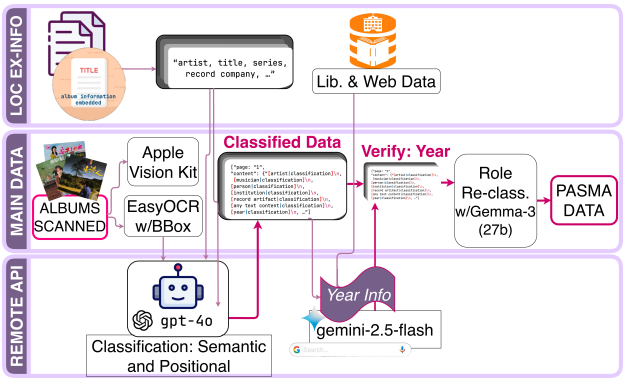


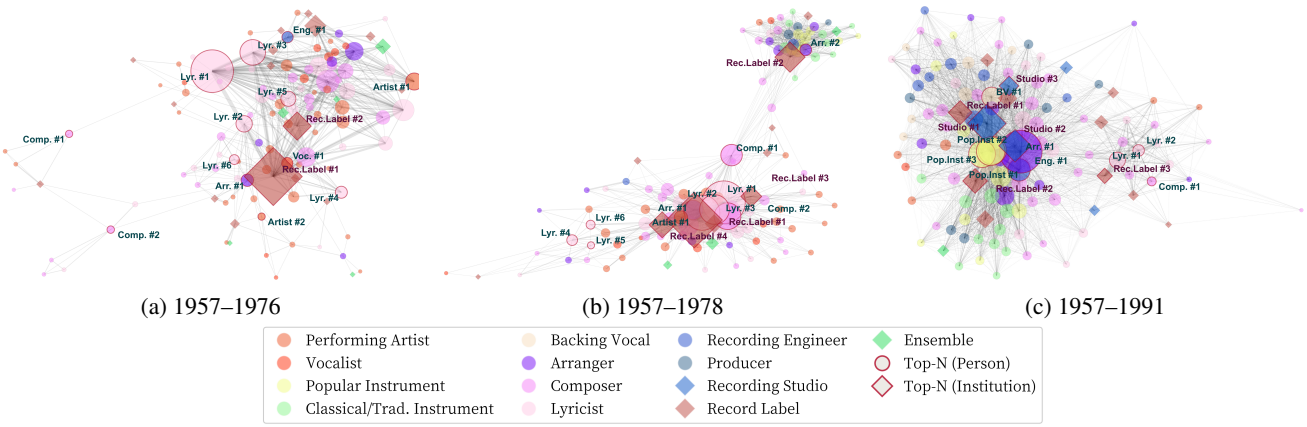
Figure 1: The PASMA data reconstruction pipeline.

3.3 Acoustic Analysis via MERT Embeddings

To examine whether the structural shifts correspond to measurable acoustic changes, we use MERT-330M [10], a foundation model pre-trained on large-scale music audio. Because PASMA does not include the recordings itself, we assembled a separate corpus of 960 tracks (1961–2016) for the acoustic analysis. Tracks were identified via the Taiwan Popular Music Database [1] and retrieved from online sources. The recordings were used solely to compute MERT embeddings for analysis; no audio is redistributed.

Historical recordings contain both front-stage vocal performance and the instrumental arrangement produced by backstage laborers. Since our interest lies in the acoustic output of the backstage network, we first apply Hybrid Demucs [20] to isolate instrumental stems, removing the singing voice. This separation is not merely a technical convenience; it is a methodological commitment consistent with our research question: we listen to what the backstage workers made, not to the stars who sang over it. The separated stems are then fed into MERT to obtain high-dimensional embeddings, which we project into a lower-dimensional space via UMAP [21] for visualization across time periods and collaboration communities.

[1] https://pmdb.taipei/



(a) 1957–1976 (b) 1957–1978 (c) 1957–1991

Figure 2: Backstage collaboration network. Color indicates role; circles are persons, diamonds institutions.

4. RESULTS

4.1 Structural Transformation of the Collaboration Network

Figures 2a–2c present the backstage collaboration network at three cumulative time points (see the legend for role color coding). The 1976 and 1978 endpoints bracket the transition identified by the five-year sliding-window metrics reported below, while 1991 marks the end of the PASMA study period.

In the first snapshot (Figure 2a, 1957–1976), the network is visibly fragmented. A main component is organized around several record labels, surrounded by smaller peripheral clusters. The bridging nodes within the main component are predominantly lyricists, while arrangers and engineers occupy marginal positions. The production ecology is centered on record labels as institutional hubs, with lyricists—rather than technical workers—serving as the primary connectors. A separate community of Taiwanese-language performers is sparsely linked to the main network, reflecting both cross-label recording prohibitions and state policies prioritizing Mandarin.

The second snapshot (Figure 2b, 1957–1978) captures a pivotal moment. Two developments converge. A separate cluster emerges, centered on a cohort of pop instrumentalists and their band's keyboardist, who also served as their arranger, all from the American military club circuit, bringing ensemble-playing experience from years of live performance. At the same time, the main component has grown considerably, drawing previously separate label-based clusters into one. Yet the military-club cohort remains partially separate, linked to the main body but not yet absorbed. The network is at a tipping point: the old label-centered ecology and the incoming studio-oriented cohort coexist but have not fully merged.

The third snapshot (Figure 2c, 1957–1991) shows the result. The network has consolidated into a single dense giant component. Recording studios—absent from the earlier graphs—now occupy the center alongside engineers and multiple arrangers. Pop instrumentalists and backing vocalists fill the core, forming a densely interconnected technical workforce. Record labels, formerly the dominant institutional hubs, have migrated to the periphery. Lyricists, who once bridged the network, are pushed to the margins. The organizing logic has shifted from label-centered to studio-centered, and the primary brokers are no longer lyricists but arrangers, engineers, and session musicians.

This visual transformation is corroborated quantitatively. Three structural metrics turn concurrently between 1976 and 1978: the clustering coefficient declines from 0.75 in 1960 to below 0.05 after 1978; the largest component ratio crosses 0.95 around 1976 and stabilizes near unity, absorbing the previously coexisting production ecologies—Mandarin-language label circuits, the suppressed Taiwanese-language community, and the military-club cohort—into a single industrial network; and Louvain modularity drops from 0.69 in the early 1970s to 0.51 by 1980. This concurrence is not built into the metrics themselves: local triadic closure, global connectedness, and community partition strength are mathematically distinct properties that need not co-vary. Their simultaneous transition at this specific window therefore constitutes evidence that 1976–1978 is not an arbitrary periodization, but a genuine structural threshold coinciding with the introduction and industrial adoption of multitrack recording in Taiwan.

4.2 Role-level Reorganization: Bridging Actors, Institutions, and Individuals

While Section 4.1 traces the network's macro-structural transformation, the question of *which actors* occupy the bridging positions driving this consolidation requires a separate analytical lens. Burt's structural hole constraint [19] identifies the specific persons and institutions whose collaborative ties span otherwise separate communities. Examined across three cumulative time points, this analysis reveals a comprehensive reorganization at three nested scales: the categorical roles that occupy bridging positions, the institutional types that anchor them, and the individual actors who embody this transformation.

Bridging Roles. Table 1 compares the role enrichment ratios among the top 30 bridging actors across three cumulative periods. The most revealing pattern is the inversion of lyricists and technical workers. In the pre-multitrack network, lyricists dominate the bridges (1.63) while arrangers are underrepresented (0.83). Adding just two years of multitrack-era production reorders the landscape: pop instrumentalists surge to 3.16, arrangers to 2.78, and producers emerge at 3.33, while lyricists fall below parity. By

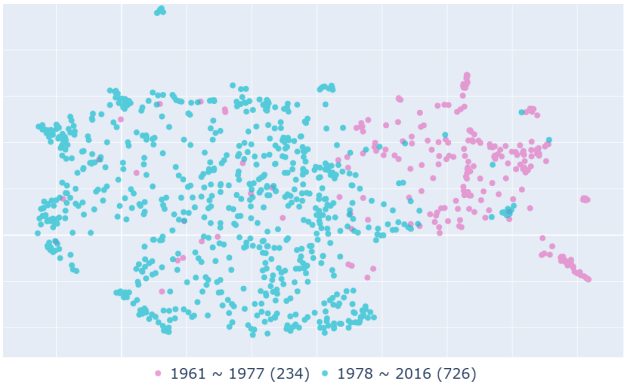
Role	’57–’76	’57–’78	’57–’91
Recording Engineer	5.83	5.00	1.57
Producer	—	3.33	—
Backing Vocal	—	—	4.01
Pop Instrumentalist	—	3.16	1.89
Arranger	0.83	2.78	3.06
Classical/Trad. Inst.	—	2.50	—
Composer	1.37	1.06	1.58
Lyricist	1.63	0.69	0.53
Vocalist	0.32	0.31	—
Performing Artist	0.61	—	—

Table 1: Role enrichment among the top 30 bridging actors across three cumulative periods. Values >1.0 indicate overrepresentation, <1.0 underrepresentation; “—”: absent from the top 30.

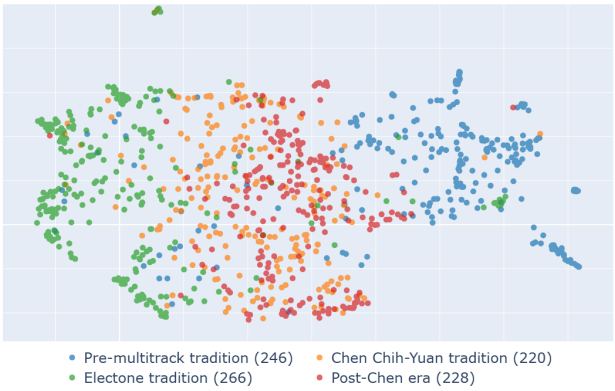
1991 the pattern stabilizes, with backing vocalists (4.01) and arrangers (3.06) as the most overrepresented bridging roles and lyricists suppressed at 0.53.

Three further patterns deserve attention. First, front-stage performers remain consistently underrepresented across all periods: vocalists hold steady below parity at 0.32 and 0.31 before disappearing from the top 30 entirely by 1991, and performing artists already register at 0.61 in the pre-multitrack network. Singing stars, in other words, were never the connective tissue of this industry—a finding that directly counters the front-stage-centric narratives our study sets out to revise. Second, classical and traditional instrumentalists rise to 2.50 in 1957–1978 but vanish from the top bridges by 1991, marking a brief transitional role: musicians with conservatory training served as cross-community connectors during the multitrack transition itself, only to be displaced by pop instrumentalists and backing vocalists once the new industrial regime stabilized. Third, recording engineers stand out differently from all other roles. Their high early enrichment (5.83, 5.00) reflects *scarcity*—just a few individuals bridging the entire network—while the drop to 1.57 by 1991 indicates not a loss of importance but the role’s *diffusion*: engineers had become numerous enough that bridging was no longer concentrated in a handful of hands.

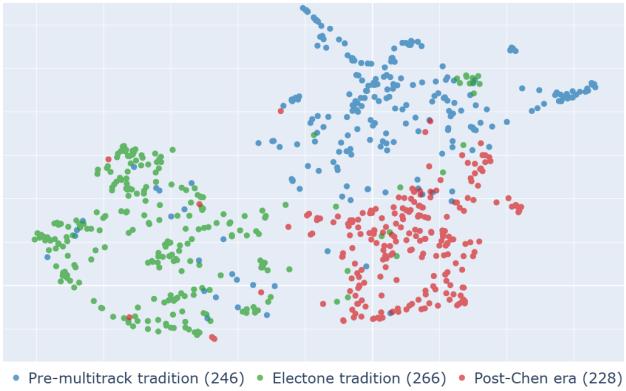
An Individual Case. The single most powerful bridge in the entire 1957–1991 network is the arranger designated as Arr. #1 in Figures 2b–2c: Chen Chih-Yuan (1941–2011). Chen first appears in the network as the keyboardist of the military-club cohort identified in Section 4.1, and transitioned around 1978 from instrumentalist to commissioned arranger. By the end of the period, he was active across virtually the entire industry—every major label, studio, and stylistic tradition in the dataset. His constraint score (0.027) places him far ahead of the second-ranked actor (0.030); he is credited on 178 albums and connects 338 distinct collaborators. As we show in the next section, this extraordinary social position has a direct counterpart in acoustic space.



(a) Colored by release period (pre- vs. post-1978).



(b) Colored by arrangement tradition.



(c) UMAP recomputed without Chen Chih-Yuan and his protégé.

Figure 3: UMAP projection of MERT embeddings from vocal-separated stems; each point is a recording.

4.3 Acoustic Analysis: MERT Embedding Results

Based on domain knowledge and the community structure identified by SNA, we defined four arrangement traditions: (1) the *Pre-multitrack tradition*, before the mid-1970s; (2) the *Chen Chih-Yuan tradition*, covering Chen and his protégé; (3) the *Electone tradition*, a parallel path rooted in the electronic organ, serving primarily the Taiwanese-language market under tighter budget constraints; and (4) the *Post-Chen era*, styles from the late 1980s onward.

Figure 3a shows the UMAP projection with recordings partitioned into pre-1978 and post-1978 groups. Despite MERT receiving no temporal information, these two groups occupy largely distinct regions of the latent space—visually consistent with the 1978 structural break identified in Section 4.2.

Figure 3b colors the same projection by the four ar-

rangement traditions. The Pre-multitrack tradition, the Electone tradition, and the Post-Chen era each occupy distinct regions of the embedding space, indicating that communities defined through collaboration patterns carry distinct acoustic signatures. However, the works by Chen Chih-Yuan and his protégé are spread across the entire space, overlapping with every other tradition. This is not noise. As shown in Section 4.2, Chen connected nearly every label, studio, and stylistic community—his omnipresence in the social network maps directly onto the sound space. To test this interpretation, we recomputed the UMAP projection with Chen's and his protégé's works removed from the input (Figure 3c). The three remaining traditions separated more sharply, with substantially less overlap. Chen's diffuse presence thus visually corresponds to his bridging role: the person who crossed all traditions also produced sound that crosses all clusters.

5. DISCUSSION

5.1 Computational Tools as Ontological Intervention

Most strikingly, the two analytical paths converge independently on the same historical watershed: SNA identifies a structural break across 1976–1978, and MERT embeddings—receiving no temporal information whatsoever—mirror this boundary in the UMAP projection around 1978. This cross-modal convergence is not a methodological coincidence; it suggests that the social reorganization of backstage labor left measurable traces in the sound itself.

This cross-modal evidence gives substance to the *ontological intervention* we proposed in Section 1. The backstage workers in our dataset—session musicians who played on dozens of albums, engineers who shaped the sound of an era, arrangers who bridged every corner of the industry—were historically present but digitally absent. By encoding these workers into the PASMA dataset, modeling their ties as a network, and extracting their sonic traces through vocal separation and MERT embeddings, we have *created* their digital existence for the first time. This intervention required the dual instruments deployed throughout our study: archival reconstruction alone would re-inscribe the silences of an industry that did not name its workers, while audio analysis alone would dissolve the historical specificity into a placeless acoustic record.

5.2 Archival Silence as Historical Evidence

A methodological objection must be addressed. One might argue that the fragmentation observed in the pre-1978 network is an artifact of incomplete data: early liner notes simply did not credit technical staff, so the network appears sparser than it actually was. This is a legitimate concern, and we do not dismiss it.

However, from a critical perspective, this archival silence is itself historical evidence. In the pre-multitrack era, recording labor was not yet professionalized or institutionalized to the degree that technical workers could demand credit. The absence of engineers and producers from early liner notes reflects not merely a documentation gap, but a real condition: these roles had not yet accumulated sufficient professional capital to claim structural recognition. The sudden, dense formalization of technical credits after 1978 captures precisely the moment when backstage intermediaries—empowered by multitrack technology and organized through studio-based workflows—consolidated enough industrial power to be named.

In SCOT terms [6], this transition from silence to visibility marks the moment when a new relevant social group achieved *closure*: the stabilization of multitrack recording as a technology was inseparable from the stabilization of the social roles that operated it. The network's structural transformation and the archival transformation are two expressions of the same underlying process. These dynamics unfolded under conditions specific to postwar Taiwan—a Cold War context that channeled American popular music through military club circuits, state policies that privileged Mandarin over local-language production, and a rapid capitalization of the recording industry during the economic boom of the 1970s and 1980s. A fuller historical account of these conditions, together with full PASMA documentation, is available on the project website. [2]

6. CONCLUSION

This study asked whether MIR's computational tools can function as instruments of ontological intervention—restoring the digital existence of historically invisible backstage laborers. Grounded in postwar Taiwanese popular music (1957–1991), we combined network analysis of the PASMA liner-note dataset with MERT embeddings of vocal-separated stems. Our findings show that the collaboration network underwent a sharp structural reorganization around 1978, and that the arrangement traditions defined by network communities occupy distinct regions in acoustic space—evidence that the social reorganization of backstage labor left measurable traces in the sound itself. In doing so, this study demonstrates how MIR's tools can serve historiographic purposes that the field's conventional applications neither anticipate nor accommodate.

Several limitations point toward future work. The PASMA dataset depends on the documentary practices of vinyl liner notes, which varied across labels and periods; future work could integrate other archival sources to address gaps in early credits. The causal relationship between multitrack technology and network centralization, while strongly suggested by the temporal evidence, cannot be fully disentangled from concurrent factors such as market growth, policy shifts, and the entry of international labels. We hope this work contributes to what Born has called *agonistic interdisciplinarity* [7]—a mode of engagement in which computational and humanistic disciplines transform each other's assumptions. If MIR is to diversify its objects and its methods, it must be willing to ask not only what algorithms can optimize, but what histories they can restore.

[2] https://bkwmt.github.io/PASMA/

7. ACKNOWLEDGMENTS

This work was supported by the Academia Sinica project "Innovative Applications of AI in Humanities and Sciences" (No. AS-IAIA-114-M06). We are deeply grateful to Pei-Ju Lin, founder of the Taipei music space *Sounds Good*, for generously providing scanned images of her private vinyl collection, without which the construction of PASMA would not have been possible.

8. AI USAGE STATEMENT

We disclose the use of LLMs in this work in two distinct capacities, in line with the ISMIR 2026 AI Usage Policy.

Methodological use. As described in Section 3 and depicted in Figure 1, the construction of the PASMA dataset relied on LLMs as integral components of a multi-stage pipeline. Because the dataset was developed over an extended period during which LLM capabilities evolved rapidly, multiple models from different families (including both commercial API-accessed models and locally-hosted open-weight models) were used at different stages of pipeline development; a single fixed model was not used throughout. All LLM-generated outputs were subject to the iterative expert validation procedure described in Section 3, and the authors take full responsibility for the resulting dataset.

Research-process use. LLMs were also used to assist the authors in the discovery, comprehension, and summarization of relevant literature, in interpretation of intermediate analytical results, and in language editing of the manuscript. All scientific claims, citations, interpretations, and conclusions presented in this paper were independently reviewed and verified by the authors, who take full responsibility for the content.

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